

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

**OFFLINE HANDWRITTEN SIGNATURE SIGNATURE FORGERY
VERIFICATION USING DEEP LEARNING METHODS**

by

Nguyen Huu Tien

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

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Supervisor:
Dr. Vu Thu Diep

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Degree Master of Software Engineering

FPT University

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Abstract

Offline signature verification is one of the most challenging tasks in biometric authentication. Despite recent advances in this field using image recognition and deep learning, there are many remaining things to be explored. The most recent technique, which is Siamese Convolutional Neural Network, has been used a lot in this field and has achieved great results. In this thesis presents an architecture that combines the power of Siamese Triplet CNN and a fully connected neural network for binary classification to automatically verify genuine and forgery signatures even if the forged signature is highly skilled. On the challenging public dataset for signature verification BHSig260, the proposed model can achieve a low False Acceptance Rate = 13.66, which is slightly better than the reference model. Based on this approach, the one-shot learning should make it possible to determine if the input image is genuine or fraudulent just from one base image. Therefore, my model is expected to be extremely suitable for practical problems, such as banking systems or mobile authentication applications, in which the amount of data for each identity is limited in quantity and variety.

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Table of Contents

Acknowledgments.....	2
Table of Contents.....	3
List of Tables	5
List of Figures	6
1 Introduction.....	1
1.1 Signature verification overview	1
1.2 Problem and Challenging.....	1
1.3 Idea and Motivation	3
1.4 Related work	3
1.5 Contribution	5
2 Dataset and Preprocessing.....	5
2.1 Dataset overview.....	5
2.2 Preprocessing	5
3 Methodology	7
3.1 Deep triplet ranking CNN architecture	7
3.1.1 <i>Siamese Neural Network(SNN)</i>	7
3.1.2 <i>Loss Function for Siamese Neural Network</i>	9
3.1.3 <i>Algorithm details</i>	11
3.2 Classification model.....	12
3.3 Final model architecture	13
3.4 Evaluation methos.....	13
4 Experiment	15
4.1 Experimental Setup.....	15
4.2 Results.....	16
5 Conclusion and Future Work	18

References	19
Copyright Acknowledgements.....	21

List of Tables

Table 1. Pairs labeling method for train and test	15
Table 2. Performance on validation set.....	16
Table 3. Performance on test set	16
Table 4. Performance on the test set with random forgeries by other user forged signatures	17
Table 5. State-of-the-art performance on BHSig260 Dataset (WD = Writer Dependent, WI = Writer Independent)	17

List of Figures

Figure 1. Bengali signature samples	5
Figure 2. Data collection and preparation flow	6
Figure 3. <i>Siamese Neural Network (SNN)</i>	7
Figure 4. Siamese Neural Networks flow illustration.....	8
Figure 5. Loss Function illustration	9
Figure 6. Triple Loss Function.....	10
Figure 7. Contrastive Loss Function	11
Figure 8. Deep Triplet Ranking CNNs	11
Figure 9. Introduce deep triplet ranking CNN architecture with Xception	12
Figure 10. Classifier model.....	12
Figure 11. Final model architecture illu.....	13
Figure 12. Deep triplet ranking CNN architecture with Xception.....	14
Figure 13. Classification flow illustration.	15

1 Introduction

1.1 Signature verification overview

A signature is a handwritten depiction of a person's name, nick name or symbol. The traditional function of signature is to permanently affix to a document, which plays a role as a physical evidence of the author's personal witness and certification of the content [1]. One of the explanations for its widespread use is that it's simple, fast, non-invasive and other people are acquainted with it in everyday life. Signature verification system aims to automatically discriminate if a biometrics sample is indeed of a claimed individual. In other words, it's wont to classify if a signature is genius or fraud. The fraud signatures, which are called forgeries, are commonly classified in three types: random (blind) forgery, simple forgery, skilled forgery. Based on the signature acquisition method, most recent signature verification systems are categorized in two types: online (dynamic) signature verification and offline (static) signature verification [2]. In the online method, the user's signature is acquired by using an acquisition device like a digitizing table. The online signatures are collected as a sequence over time and contain numerous information, such as pen position, pen inclination, pressure, etc. On the other hand, signatures acquired from the offline method are the digital images of the user's signatures signed in the document after the writing process has completed. When the online method can contribute more diverse features for the verified process, it requires significant devices and techniques to acquire data, which makes the system more expensive and cumbersome. On the other hand, the data from offline methods are easier to obtain, and the image digital signatures are more suitable for practical problems. However, the lack of informative features compared to the other method and the quality instability of image data make the verification process more challenging.

1.2 Problem and Challenging

The problem of offline signature verification is often modeled as follows: given a group of genuine signatures of users, a model is trained to extract meaningful features from them. Then, the model is employed for verification: a user claims their identity and provides one or some new signatures, which can be employed by the model to classify those signatures as genuine (belong to the claimed individual) or forgery (created by someone else).

In signature verification, one of the most important and familiar challenges for this task is that handwritten signatures have high intra-class variability, which means there are always differences between each handwritten signature from the same person [2]. This issue leads to an aggravation when considering skilled forgeries, especially with the presence of low inter-class variability in the dataset. In short, when there are large differences between each signature sample (high intra-class variability) and the forgeries are very skillful that they are nearly the same with the genuine (low inter-class variability), the possibility of confusing the forgeries with the genuine is higher.

The second challenge when training an automatic signature verification system comes from the presence of partial knowledge during training. In the realistic scenario, we've got only to access genuine signatures for users enrolled within the system during the training phase, in line with [3]. During operations, however, we would like the system to not only be able to accept the real, but also to reject forgeries. To solve that, the forgery signatures for each user should be required for a better classification. But in general, it is not reasonable to require users to provide signatures that forge their own. Even if they can be collected and created from the service provider or from a third party, it would be a challenge to create a good forgeries data which is good enough to be absolutely distinguishable from the genuine signatures.

Thirdly, the amount of training data is always being considered. During the enrollment process in the real application, users are often required to produce only some samples of their signatures. In the meantime, some approaches like the Writer-Independent approach need a large enough amount of data to perform well. That's why it's hard to create an efficient system with the very limited data collected from enrolled users in a very real case scenario. Whether or not there's an outsized number of users supplying their signatures to the system during the training phase, the performance of the classifier has to be superb for the new user, for whom supply only few samples of signatures.

Last but not least, like the face recognition system, an effective real-world signature verification system should be able to deal with the one-shot learning problem. Which means the verification application should instantly classify the genuine and fraudulent signatures of a user from just one genuine signature of that individual. However, deep learning algorithms do not work well with a very small dataset, especially if you only have one training example.

1.3 Idea and Motivation

Last but not least, like the face recognition system, an effective real-world signature verification system should be able to deal with the one-shot learning problem. Which means the verification application should instantly classify the genuine and fraudulent signatures of a user from just one genuine signature of that individual. However, deep learning algorithms do not work well with a very small dataset, especially if you only have one training example.

In general, signature verification systems have two phases: feature extraction and classification. In the first phase, the feature obtained from the extraction must be meaningful enough to represent the distinctive features of each individual (Writer-Dependent approaches) or show the clear difference between the genuine and forged signatures (Writer-Independent approaches). Then, based on those features, a classifier will be used to make the final prediction.

1.4 Related work

In 2008, Impedovo et al. [1] provided a summary in automatic signature verification, with specific attention to most outstanding advancements in machine learning. Then, Shah et al. [2] presented a survey about the critical evaluation of 15 techniques applied on offline signature verification systems, which classify each work according to the feature extraction methods, classifiers and overall strengths and limitations of the systems. However, these reviews don't update to capture more recent trends, specifically the usage of deep learning methods, which have demonstrated superior results in multiple benchmarks.

Comprehensive surveys and state of the art reviews of the recent literature are often found within the works of [3, 4]. In step with these reviews, most deep learning methods for this task can archive results with high accuracy on an outsized dataset. Beside, among numerous researchers, a growing number of researchers have realized that "learning with small datasets" are a key factor for the success of signature verification in real practical scenarios. Authors like Bouamra et al. [5] attempt to train the model with atiny low number of signature samples through a one-class support vector machine (OC-SVM) classifier. Hafemann et al. [6] propose an answer supported a meta-learning approach.

In recent years, many interesting techniques don't depend on hand-engineered feature extraction anymore. With the increase of the many deep learning approaches, the power to execute more

meaningful feature representation from data (like pixels, just in case of images) by itself can promote faster and efficient learning, which helps researchers to save lots of significant amounts of work. In this field, Khalajzadeh et al. [7] applied CNNs for Persian signature verification, but only considered random forgeries in their tests.

Koch et al. [8] firstly propose an approach about Siamese network and one-shot learning for image recognition. This approach outperformed many available methods by previous authors. Moreover, they have argued that this approach can extend in other domains, especially for image classification.

Schoff et al. [9] came up with the same idea of applying the Siamese network, but in a different way. In this research, they propose a new system called FaceNet, which learns a way to map face images to a compact Euclidean space where distances directly correspond to a measure of face similarity. They achieved high accuracy ($95.12\% \pm 0.39$) in their published dataset, and also introduced new concepts about harmonic embeddings and harmonic triplet loss to deal with the face classification task.

The above approaches [8, 9] both apply Siamese structure including deep CNNs for getting a new representation of image feature, which will be used to classify tasks by comparing them over their distance between each other in new vector space. Those techniques are very flexible and do not need a dataset with millions of samples to make them work.

Dey et al. [10] applied Convolutional Siamese network with contrastive loss function for writer independent offline signature verification and their system performs very well on cross dataset for this task. In addition, their approach has surpassed the state-of-the-art result on most of the benchmark Signature datasets, which is encouraging this technique for further research. However, when performing evaluation across different datasets, the accuracy of their system gradually decreases on the datasets which have more distinctive features compared to the training dataset.

Chhabra et al. [11] have constructed an interesting method based on the Convolutional Siamese Net architecture from FaceNet [9] and triplet loss concept to solve the one-shot problem for off-line signature verification. Their model has high accuracy and generalizability on the public signature database. However, they do not show the evaluation result in various databases and the performance comparison with other state-of-the-art methods.

1.5 Contribution

Inspired by the work of Chhabra et al. [11], this study improves the network architecture of Deep Triplet Ranking CNN Network by applying more robust transfer learning models and more efficient classification methods. The work focuses mainly on forgery detection in signature verification systems with one-shot learning strategy, which means our model may not perform well on recognition identity from signature but to eliminate forgeries as much as possible. The model is evaluated on the dataset BHSig260, the challenging public dataset for signature verification.

2 Dataset and Preprocessing

2.1 Dataset overview

In this work, the signature dataset BHSig260 is used. These data are public, free to access, and contain skilled forgeries. BHSig260 dataset contains the offline signatures of 260 persons, in which 100 persons were signed in Bengali and the remaining were signed in Hindi. For each of the signers, 24 genuine and 30 forged signatures are available. This results in 2400 genuine signatures and 3000 forged signatures in Bengali, and 3840 genuine and 4800 forged signatures in Hindi. My experiment is only in the Bengali signatures datas from this dataset.

টিয়া মল্লিক	সুজিতা মল্লিক
সুজিতা মল্লিক	সুজিতা মল্লিক
সুজিতা মল্লিক	সুজিতা মল্লিক

Figure 1. Bengali signature samples

2.2 Preprocessing

Through analyzing the characteristics of the signature image, we have carried out the following preprocessing steps to improve the efficiency of the recognition process:

- The color factor is not important when comparing the dissimilarity of genuine and fraudulent signatures. That is why we convert all signature's image to grayscale.
- Using a median filter with kernel size 3x3, most salt and pepper noise are easily eliminated.
- After removing noise, using the Otsu threshold to convert the gray images to be the binary image. In addition, invert to the negative image so the background become zero and only the signature's area has pixel value 255.
- Finally, resizing all signature images to a fixed size for feeding into the network 128x128.

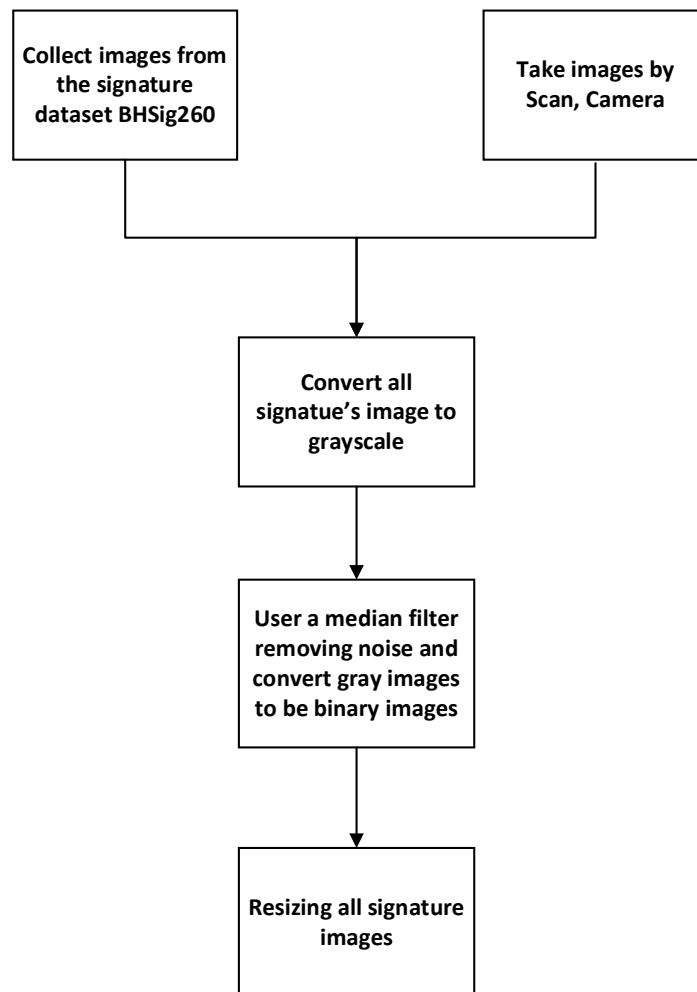


Figure 2. Data collection and preparation flow

3 Methodology

3.1 Deep triplet ranking CNN architecture

3.1.1 Siamese Neural Network(SNN)

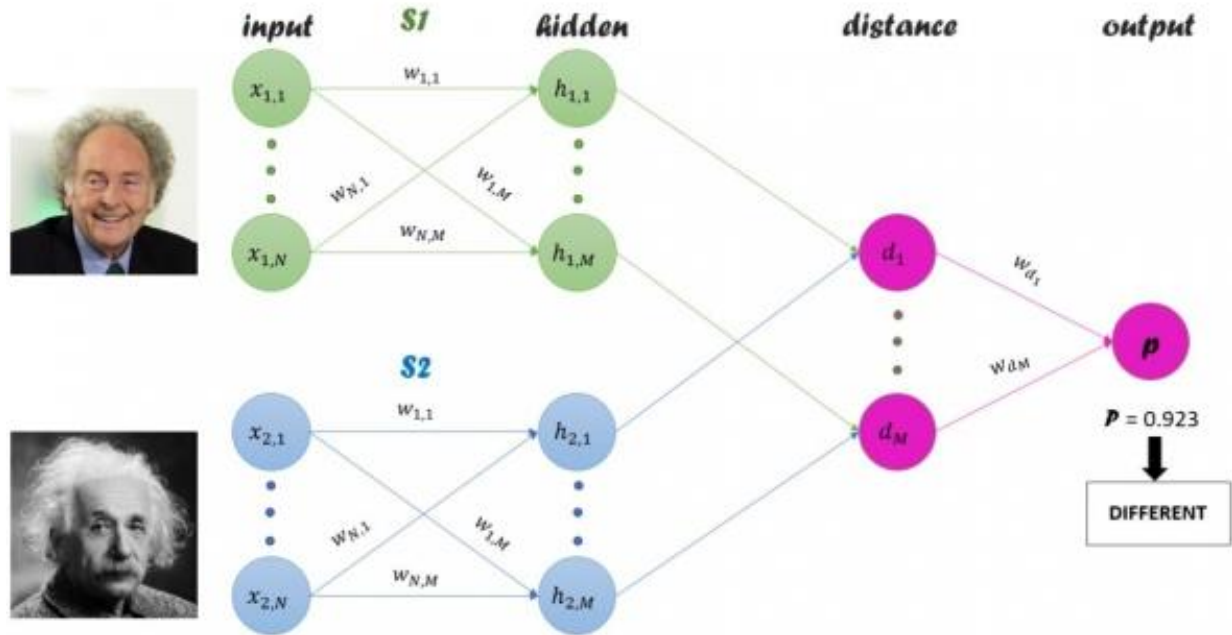


Figure 3. Siamese Neural Network (SNN)

A *siamese neural network* (SNN) is a class of neural network architectures that contain two or more identical sub-networks. “Identical” here means they have the same configuration with the same parameters and weights. Parameter updating is mirrored across both sub-networks and it’s used to find similarities between inputs by comparing its feature vectors. These networks are used in many applications. As siamese networks are mostly used in verification systems (face recognition, signature verification, etc.)

Traditionally, a neural network learns to predict multiple classes. This poses a problem when we need to add or remove new classes to the data. In this case, we have to update the neural network and retrain it on the whole data set. Also, deep neural networks need a large volume of data on which to train. SNNs, on the other hand, learn a similarity function. Thus, we can train the SNN to see if two images are the same (which I’ll demonstrate below). This process enables us to classify new classes of data without retraining the network.

The SNN works as follows:

- Select a pair of input data from the dataset.
- Pass each image through each SNN's Sub-network for processing. The output of the Sub-networks is a Embedding vector.
- Calculate Euclidean distance between two Embedding vectors.
- A *Sigmoid Function* can be applied over the distance to give the score value in the segment $[0,1]$, shows the degree of similarity between two Embedding vectors.

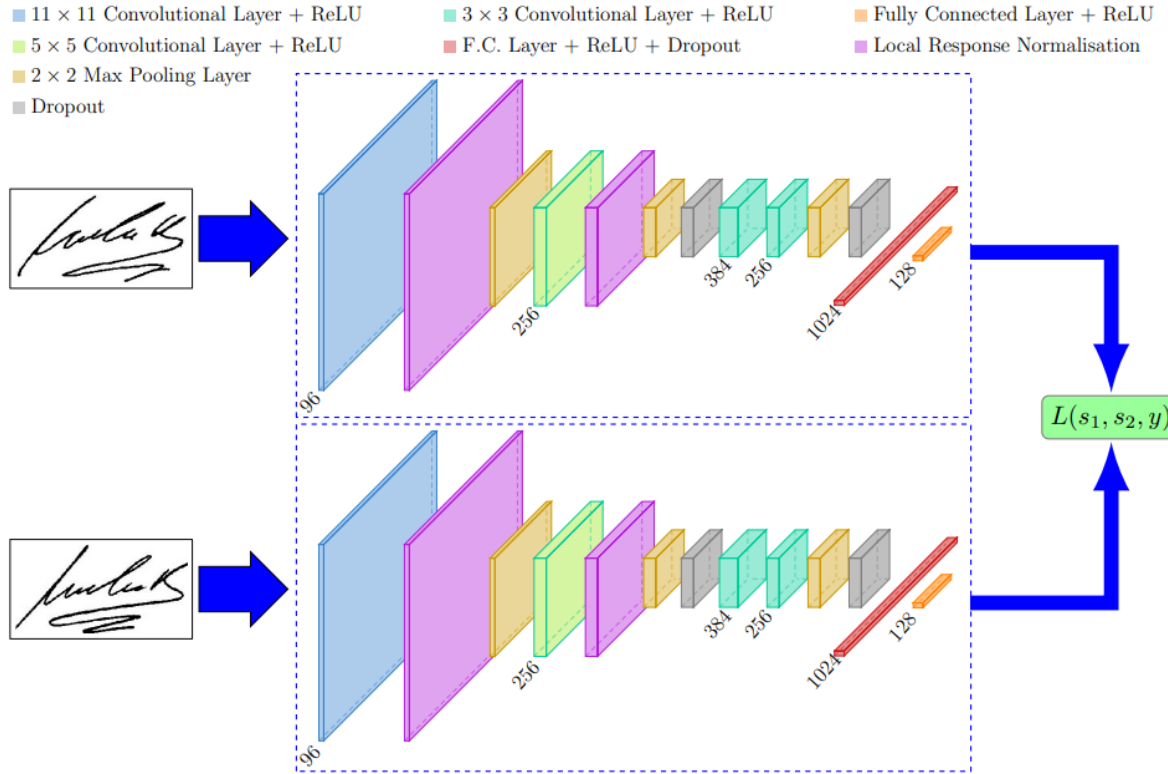


Figure 4. Siamese Neural Networks flow illustration

Advantages of SNN

- The amount of data required to train the SNN is very small. Just a few samples is enough (1-5 samples) to train the SNN. The method it uses here is *One-Shot Learning* or *Few-Shot Learning*. Because we need so little training data, we don't worry about data being imbalanced(*Image Imbalance*).
- The ability to combine with other classifiers is high. Since the learning mechanism of SNN is different from other common classifiers, we can completely combine them together. This usually produces better results.

- Learning from semantic similarity: SNN focuses on learning features in deeper layers, where similar features are placed close together. Therefore, it can partly understand the semantic similarity of input data.

Disadvantages of SNN

- Longer training times. SNN learns in pairs, so its learning ability is slower than other Neural Networks.
- Do not show the probability of each class in the output. SNN only gives one score value in the interval [0,1], showing the similarity between 2 input data. The closer the score is to 1, the more similar the 2 input data are and vice versa.

3.1.2 *Loss Function for Siamese Neural Network*



Figure 5. Loss Function illustration

Because the SNN learns in a pairwise fashion of input data, the Cross Entropy Loss Function is not usually used. Instead, Two Loss Functions are Triple Loss and Contrastive Loss, which are used more.

Triple Loss Function

The idea of Triple Loss is to use triple input data including: Anchor (A), Positive (P) and Negative (N). The positions that the distance from A to P is minimized, while the distance from A to N is maximized during model training.

$$L(A, P, N) = \max(\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 + \alpha, 0)$$

In the above formula

- α is *margin*, used to emphasize the difference or similarity between pairs of input data.
- $f(A), f(P), f(N)$ are the feature vectors of the input data A, P, N, corresponding.

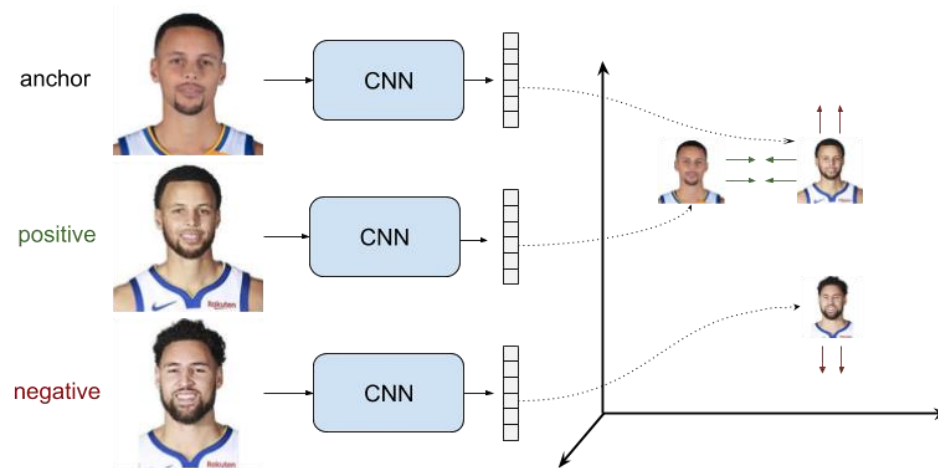


Figure 6. Triplet Loss Function illustration

Contrastive Loss Function

The idea of Contrastive Loss is similar to Triplet Loss, the difference is that Contrastive Loss only uses one pair of input data, either of the same types or different types. If they are of the same type, the distance between their feature vectors will be minimized, and if they are different, the distance between their feature vectors will be maximized during training.

Formular of *Contrastive Loss*:

$$(1 - Y)\frac{1}{2}(D_w)^2 + (Y)\frac{1}{2}\max(0, m - D_w)^2$$

In there, D_w is the distance Euclidean:

$$\sqrt{(G_w(X_1) - G_w(X_2))^2}$$

G_w is the output of the SNN for an input data.

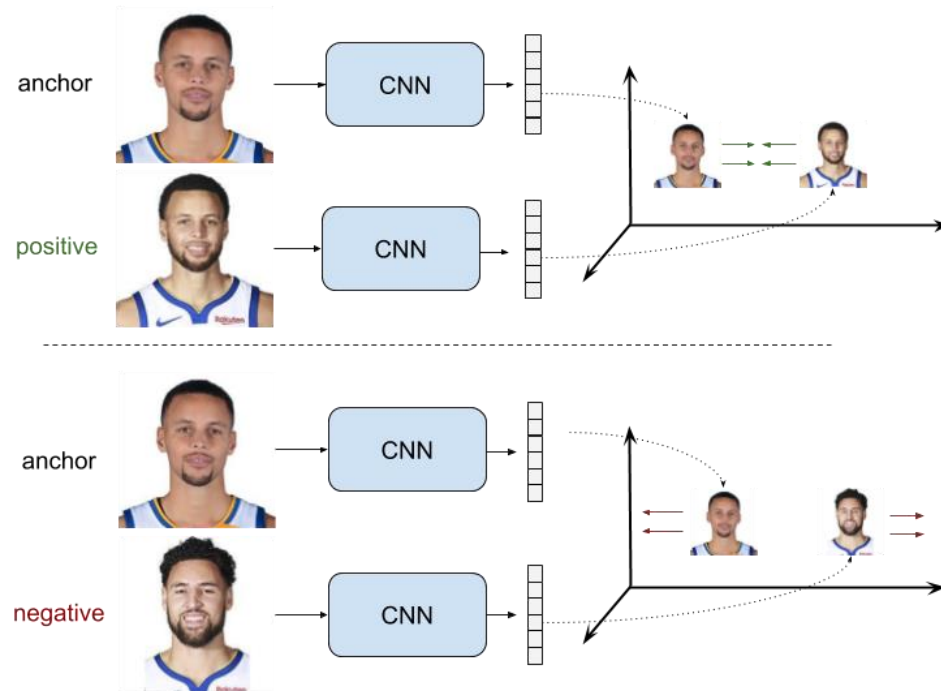


Figure 7. Contrastive Loss Function illustration

3.1.3 Algorithm details

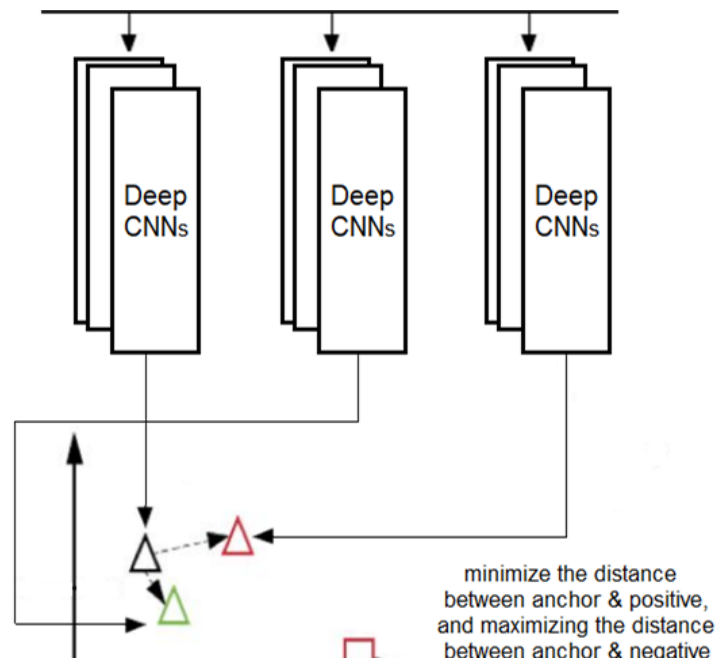


Figure 8. Deep Triplet Ranking CNNs

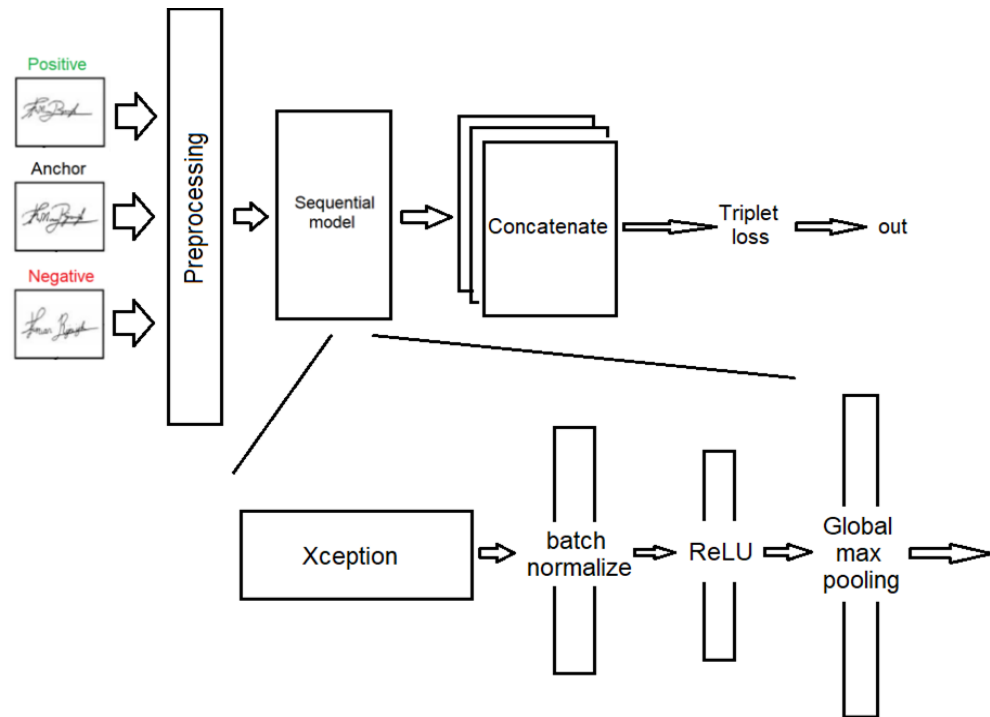


Figure 9. Introduce deep triplet ranking CNN architecture with Xception .

3.1.4 Classification model

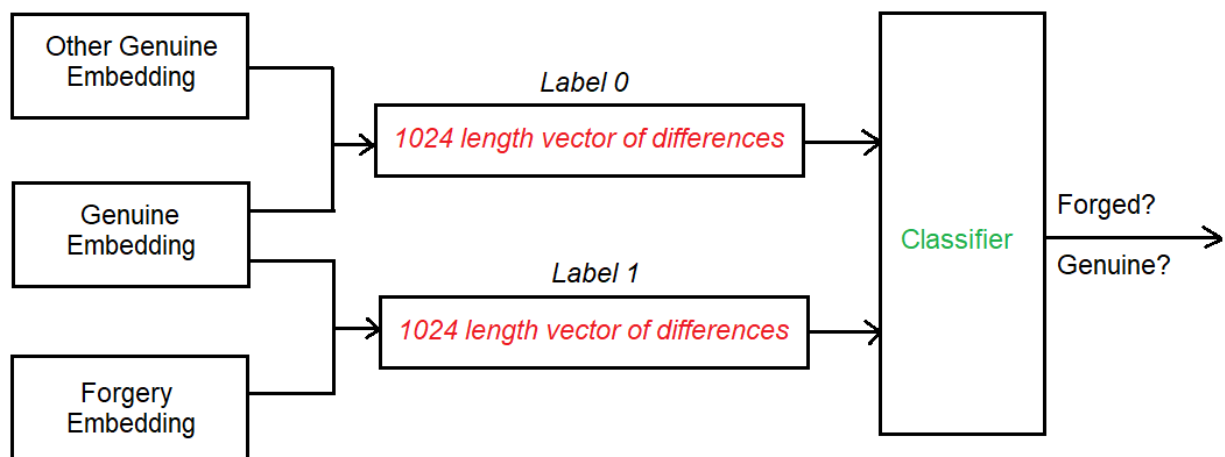


Figure 10. Classifier model illustration

3.1.5 Final model architecture

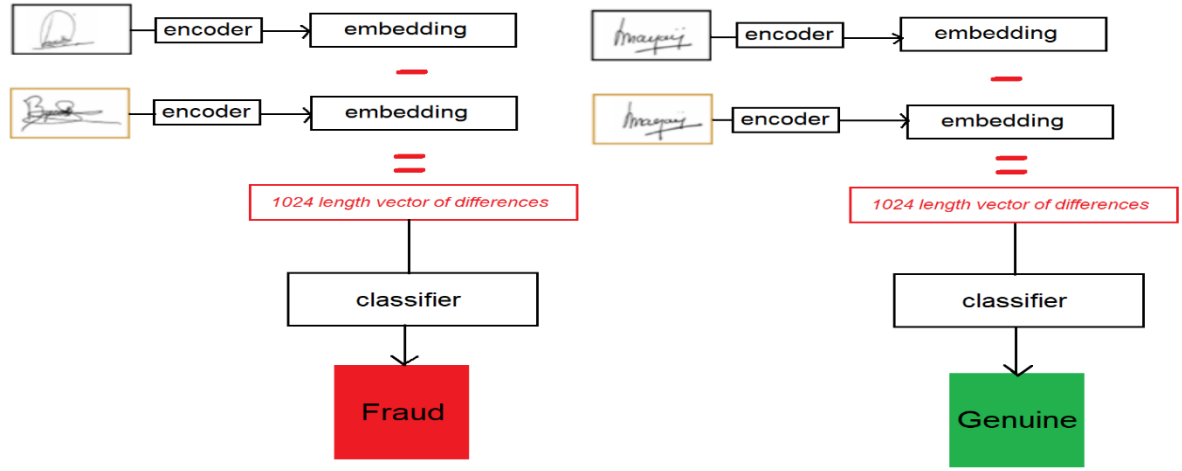


Figure 11. Final model architecture illustration

3.2 Evaluation methos

The architecture proposed in Figure 12. is modified from the architecture introduced by Chhabra et al. [11]. We uses 3 sysmetrics deep Convolutional Neural Networks played as the subnetworks in the Siamese architecture. *Instead of building the new deep CNNs and training it from scratch, the Xception model, a pre-trained model in ImageNet is used for three branches.* The transfer-learning strategy is used to reduce time for the training process: firstly, the model with the weights trained on ImageNet dataset will be removed the fully-connected layers on top, then all layers are freezed except the exit flow for train on the new prepared dataset. This strategy helps our model have the ability to extract the high feature representation of the image (from the freezed layers) while still being able to learn the important features from the signature image in new dataset. In this case, the Xception was chosen because of its superiority over models VGG-16, ResNet and Inception V3 and has been summarized in [12]. Moreover, it has fewer parameters than Inception V3, which can save more memories on the training and deployment processes.

In this design, our model accepts the input as three signature images categorized as: anchor (an individual's genuine signature), positive (another genuine signature of the same individual) and negative (a forged signature of the same individual). After the preprocessing, those triplet images are fed through the triple subnetworks in which each is constructed by transfer-learning Xception

top up with a stack of a Batch Normalization, ReLU activation function and a global max pooling layer.

After being concatenated, our network uses a Lambda layer to compute two distances between embedding vectors: “positive-anchor”, “anchor-negative”. And the last calculates the triplet loss. Of course triplet loss optimization will continue to update the weights of the sequential part so that the embedding images have adjusted distance to become closer if they are genuine, or become further away from the forgeries.

In the next phase of genuine-forgery classification, we continue to use the idea of using distance vectors for the classifiers of weighted authors in [11] (Figure 13.), but replace the logistic regression classifier with a new classifier stack. The new binary classification includes a fully-connected Layer and ReLU activation function, which is top-up with sigmoid function to calculate the probability that a signature is forged or genuine.

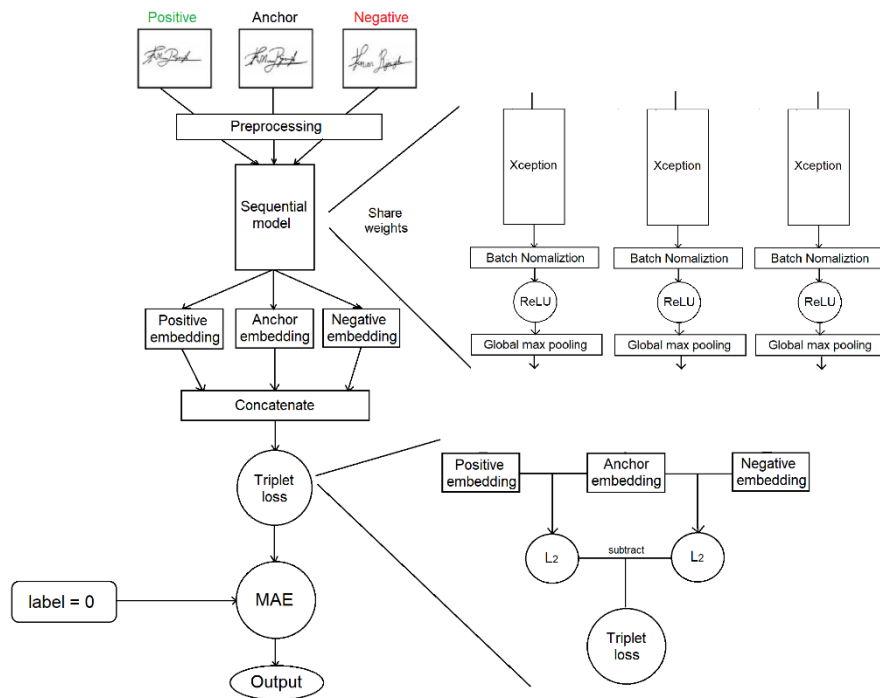


Figure 12. Deep triplet ranking CNN architecture with Xception.

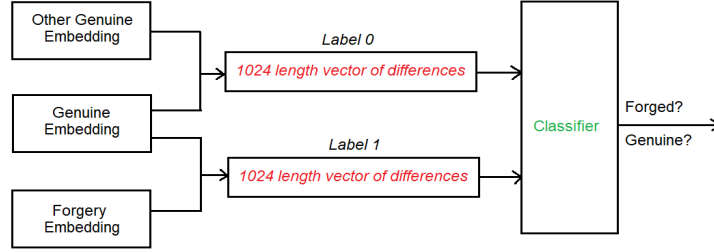


Figure 13. Classification flow illustration.

4 Experiment

4.1 Experimental Setup

This study uses google colab for experiment.

First of all, I split 20% of users and take their signature images because the test data. When forming the triplet combinations to train the Triplet net, we take an anchor image (genuine signature of a person) and placing it in conjunction with both a positive sample (another genuine signature of the identical person) and a negative sample (a forged signature by some other person of the identical person). By that strategy, from dataset BHSig260 Bengali, we have 654120 triplet combinations for training. The training pairs for the classifier are formed after step of embedding of all signatures using the train triplet net. I have the vector pairs, label them (see Table 1) and feed them to the classifier.

Table 1. Pairs labeling method for train and test

Set(Train/Test)	Pair	Label
Train Users	Genuine - Genuine	0
	Genuine - Forgery	1
Test Users	Base Genuine - Genuine	0
	Base Genuine - Forgery	1

4.2 Results

The triplet net is trained by using Adam Optimizer with the learning rate equal to $1e-5$ and mean absolute error for backpropagation to get the final encoding. The triplet combination is randomly split into a train set and validation set with ratio 2:1, respectively.

In this work, the positive class is the forged signature and the negative class is the genuine signature. We assess or compare the performance of system based on the following values:

- Area Under the Curve (AUC): the measure of the power of a classifier to differentiate between classes.
- False Acceptance Rate (FAR): the percentage of identification instances within which unauthorised persons are incorrectly accepted. $FAR = FNR = FN / (FN + TP) = 1 - \text{recall}$. This parameter is that the most important during this study, the smaller the better.
- False Rejection Rate (FRR): the percentage of identification instances within which authorised persons are incorrectly rejected. $FRR = FPR = FP / (FP + TN)$.
- Equal Error Rate (EER): The intersect of lines FAR, FRR.

The Table 2 and Table 3 show the performance of the final model for detecting forgery and genuine signatures:

Table 2. Performance on validation set

AUC (%)	ERR	FAR	FRR
99.99	0.13	0.00	0.00

Table 3. Performance on test set

AUC (%)	ERR	FAR	FRR
86.16	22.75	18.57	27.97

Beside, we have the performance on the test set in case of the pair Base Genuine - Forgeries is no longer depend on identity anymore, which means we construct the pair from the base genuine signature with all forgery signature in the dataset to get more positive label (the forged signature) for testing (Table 4)

Table 4. Performance on the test set with random forgeries by other user forged signatures

AUC (%)	ERR	FAR	FRR
99.34	14.18	13.66	27.97

The result from Table 4 can be used to compare with some state-of-the-art and the recent studies with Siamese Network in the following Table 5:

Table 5. State-of-the-art performance on BHSig260 Dataset (WD = Writer Dependent, WI = Writer Independent)

Language	Type	Features & algorithm	FRR	FAR
Bengali	WI [10]	SigNet	13.89	13.89
	WI [13]	Dutta et al	14.43	15.78
	WI [14]	Pal et al.	33.82	33.82
	-	Our model	27.97	13.66

When the solution is used in real applications, only one genuine signature image is necessary for new user. The image will be preprocessed and feed through the encoder to get the base embedding vector. This vector will be stored in the database and be marked by an id to query in future.

When a new signature image is attempted on the system, if the owner of the new signature claims to be anyone in the system by promoting the id, the system will use that id to track and query the base embedding out for comparison. The system then uses the encoder to get the embedding of the new signature image, and get the vector's difference between the embedding of the base signature and the new signature. Lastly, the classifier will use this vectors difference as an input to predict the probability if the new image is a forged signature or not. The higher the value of this predicted result, the higher the probability that the new signature is a fake one.

5 Conclusion and Future Work

The thesis proposes an effective method for verifying the offline handwritten forgery signatures based on the improvement of a highly applicable study. The article has performed image preprocessing to improve recognition efficiency, improve in processing steps such as using transfer learning strategy and choosing a suitable machine learning architecture model for the problem.

From the result, I can achieve a good FAR result on the BHSig260 dataset which is slightly better than the reference results.

In the future, the study need to find more suitable preprocessing methods for different kinds of datasets. The model also need to be tuned hyperparameters of more powerful triplet ranking loss, and may need to apply triplet online mining techniques to let the learning process be better.

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